

tion. Fuel switch has received impetus through blending mandates and a focus on commercial affordable fuel cells. While wide adoption of the latter may take a few years away, electrification of transport and logistics by expanding our railway network offers a viable alternative today.

This Budget emphasises the implementation of three main railway corridors under the PM Gati Shakti programme: one dedicated to energy, minerals and cement; one dedicated to port connectivity; and one to cater to high traffic density corridors. The corridors will improve efficiencies and will replace road transport to some degree. In cities, there is an emphasis on expanding metro rails and promoting faster adoption of e-vehicles for mass transit.

We must adopt a multi-pronged approach to promote clean transportation and logistics—one that will focus on developing multiple clean transport and logistics options as well as boost innovation and production of affordable e-vehicles.

Materials Transformation

Circle Economy, a think tank, estimates that the world consumes around 100 billion tonnes of materials and recycles approximately 8.6%. And according to the Ministry of Electronics and Information Technology (MeitY), India alone is expected to consume about 15 billion tonnes of materials by 2030. Thus, mining, processing and use of virgin materials poses a significant strain on the environment.

India's response has been to focus on efficient waste management and extended producer responsibilities to recycle certain materials such as plastics. Transforming material systems, however, is challenging as it requires significant investments and innovations in alternative materials, green chemistry and business models based on re-use/re-manufacture as a core competency.

This Budget made one of the first real attempts to accelerate invest-

ments in alternative regenerative materials as it promotes bio manufacturing and refineries. But more is needed.

Transformation of Food Systems

Food systems are responsible for around 18% of the greenhouse gas emissions in India. We need to balance our need to achieve food security and transform food systems at the same time. This will require a focus on types and use of fertilisers. In the Budget, the Finance Minister called for greater use of sustainable fertilisers that offer higher efficiency when used in lower quantities.

Continuous investments are required to improve yields, adopt regenerative agricultural practices, increase mechanisation, and promote responsible use of agricultural waste.

Our efforts to mitigate climate change must be backed by efforts on adaptation. We have already crossed

the safe threshold of climate change and the effects are visible: raging wildfires, erratic monsoons and frequent cyclones. We need our socio-economic systems to adapt. While the Budget touches upon the importance of integrating coastal commerce to promote climate resilience, we do not have a formal policy or a uniform approach towards integrating climate adaptation in infrastructure planning and operations.

Climate solutions need to be affordable to see large-scale adoption. Creating scale, especially in novel technologies, will require both market and non-market interventions.

The Interim Budget was noteworthy for two reasons: its brevity, and the vision of Viksit Bharat by 2047. By coupling economic growth with social justice and climate actions, India can usher in an era of responsible, inclusive and sustainable growth. **BT**

The author is Chairperson, PwC in India. Views are personal.

